

Table 1: No drift scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$	EO $\downarrow$
Aranyani	0.8538 ± 0.0041	0.0860 ± 0.0026	0.0467 ± 0.0083
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000	0.0714 ± 0.0000
RFR	0.8050 ± 0.0079	0.0585 ± 0.0073	0.1119 ± 0.0382

Formatting Instructions For NeurIPS 2025

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Abstract

1

2 1 Introduction

3 In the high stakes world of algorithmic decision making, the outcomes of machine learning models
 4 can have significant impacts on individuals and society in areas such as healthcare, criminal justice,
 5 lending, hiring. In this contexts, fairness, is not just a desirable property but a critical requirement
 6 TODO: [ADD CITATIONS]. In particular, group fairness, which requires predictive behaviour
 7 remains equitable across demographically defined groups, has emerged as a key focus of research in
 8 this area.

9 Notwithstanding the significant progress in developing group fairness algorithms, most of the existing
 10 work assumes a stationary data distribution which, in the real world, rarely holds. Instead, data
 11 distributions evolve over time: policy shifts, seasonal effects and changes in societal norms can lead
 12 to distribution shifts (often referred to as concept drift) that can undermine predictive performance
 13 and fairness guarantees.

14 While online learning algorithms have been created to preserve predictive accuracy, there has been
 15 less work on how to preserve fairness under concept drift. This is a critical gap, because fairness
 16 violations may remain undetected longer than accuracy degradation, disrupting the trustworthiness of
 17 machine learning systems and leading to significant harms.

18 To address this challenge it is important not only to detect distributional changes as they occur but, at
 19 the same time, characterize their impact on fairness and react accordingly. In this paper, we study
 20 fairness-aware learning under concept drift in online settings and investigate how a novel framework
 21 for detecting and responding to concept drift can be used to preserve fairness guarantees over time.

22 2 Experiments

23 Tables 1–5 report per-scenario results on Adult with mean $\pm$ standard deviation across seeds from
 24 files/experiments. Aranyani results use 4 available seeds; ARF and RFR use 5 seeds.

Table 2: Abrupt gender scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$	EO $\downarrow$
Aranyani	0.8607 ± 0.0199	0.0414 ± 0.0192	0.0832 ± 0.0614
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000	0.0714 ± 0.0000
RFR	0.8030 ± 0.0045	0.0622 ± 0.0042	0.0857 ± 0.0284

Table 3: Gradual gender scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	0.8573 ± 0.0026	0.0129 ± 0.0073
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000
RFR	0.8040 ± 0.0074	0.0690 ± 0.0118

References

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- [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauero, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.
- [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural Simulation System*. New York: TELOS/Springer-Verlag.
- [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

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Table 4: Gender-relationship decouple scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$	EO $\downarrow$
Aranyani	0.8516 ± 0.0052	0.0854 ± 0.0065	0.0447 ± 0.0174
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000	0.0714 ± 0.0000
RFR	0.7890 ± 0.0096	0.0713 ± 0.0142	0.1214 ± 0.0053

Table 5: Occupation gender reversal scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$	EO $\downarrow$
Aranyani	0.8516 ± 0.0056	0.0524 ± 0.0060	0.0467 ± 0.0122
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000	0.0714 ± 0.0000
RFR	0.7990 ± 0.0124	0.0728 ± 0.0057	0.1262 ± 0.0247

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